



7313 - Determining the Origin of the Gas in the 49 Ceti Debris Disk

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) -49-Cet
	2		NIRSpec IFU Spectroscopy	(3) -48-Cet-PSF-reference

ABSTRACT

Millimeter observations of debris disks have revealed that many are surprisingly gas rich in molecular CO. The origin of this gas is debated, as it could be leftover primordial gas from the parent protoplanetary disk, or it could be of a secondary origin where the destruction and outgassing of volatile rich minor bodies releases CO into the circumstellar environment. Answering this question on the gas origin is key to understanding this late stage of planet formation. The key prediction that distinguishes primordial versus secondary gas is that primordial gas is dominated compositionally by H₂, which can shield the CO to the age of the system if it is dense enough in the disk surface. 49 Ceti is a 40 Myr old star that is host to a large

reservoir of CO detected with ALMA; however, the ALMA observations have not determined the origin of the gas in this system. In Cycle 1, astronomers used the NIRSpec IFU PRISM mode to search for water ice in this system, but surprisingly discovered fluorescent CO emission that traces the upper most surface level of this gas disk. Here we propose to follow up this discovery but at the highest NIRSpec IFU spectral resolution. This will enable the measurement of each individual ro-vibrational line to determine the level populations of each rotational level of CO in the disk surface layer. We will use the rotational level populations to determine the H₂ density in the disk surface with a non-LTE model, thus determining the origin of the gas. This proposal will serve as a pathfinder that demonstrates that the NIRSpec IFU can determine the H₂ content in these disks surfaces and ultimately unveil the origin of the gas in debris disks.

OBSERVING DESCRIPTION

We propose to obtain NIRSpec IFU G395H mode observations of the 49 Ceti debris disk to follow up the discovery of fluorescent CO emission lines from the NIRSpec IFU PRISM mode in Cycle 1. The higher spectral resolution data will enable the measurement of the fractional rotational populations of CO that is present in the upper most disk surface layer. This measurement, along with a non-LTE model, will constrain the H₂ density in the disk surface layer which will in turn reveal if the H₂ content is large enough to shield the gas to the 40 Myr age of the system. If the H₂ density is not large enough, then this will favor a secondary origin of the CO from the release of volatiles from minor bodies in the system. This program will serve as a pathfinder in demonstrating that fluorescent CO emission detected with NIRSpec can be used to constrain the H₂ density in these gas-rich disk surfaces for the first time and ultimately determine if the gas is primordial or of secondary origin.

We will observe a PSF reference star immediately after 49 Ceti in order to facilitate PSF subtraction and remove the stellar PSF. This will be a necessary step in detecting the faint CO emission lines next to the bright host star. We will include a 9-point dither pattern to best reconstruct the PSF for the optimal PSF subtraction performance.

Proposal 7313 - Targets - Determining the Origin of the Gas in the 49 Ceti Debris Disk

	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
Fixed Targets	(1)	-49-Cet	RA: 01 34 37.7787 (23.6574112d) Dec: -15 40 34.90 (-15.67636d) Equinox: J2000	Proper Motion RA: 94.351 mas/yr Proper Motion Dec: -3.1299999136535916 mas/yr Parallax: 0.0174725" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar gas, Debris disks] Extended=YES				
Fixed Targets	(3)	-48-Cet-PSF-reference	RA: 01 29 36.1357 (22.4005654d) Dec: -21 37 45.61 (-21.62934d) Equinox: J2000	Proper Motion RA: 56.468999999999994 mas/yr Proper Motion Dec: 4.134 mas/yr Parallax: 0.013896499999999999" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[A stars]				

Proposal 7313 - Observation 1 - Determining the Origin of the Gas in the 49 Ceti Debris Disk

Observation	Proposal 7313, Observation 1											Thu Oct 16 17:00:18 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	-49-Cet	RA: 01 34 37.7787 (23.6574112d) Dec: -15 40 34.90 (-15.67636d) Equinox: J2000				Proper Motion RA: 94.351 mas/yr Proper Motion Dec: -3.1299999136535916 mas/yr Parallax: 0.0174725" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar gas, Debris disks] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	0.0		-16.7		0.0		10.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	5	false	true	NONE	9	45	3939.0	

Proposal 7313 - Observation 1 - Determining the Origin of the Gas in the 49 Ceti Debris Disk

Special Requirements	Aperture PA Range 201.2 to 221.2 Degrees (V3 62.22835083 to 82.22835083) Group Observations 1, 2, Non-interruptible
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Proposal 7313 - Observation 2 - Determining the Origin of the Gas in the 49 Ceti Debris Disk

Observation	Proposal 7313, Observation 2												Thu Oct 16 17:00:18 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	-48-Cet-PSF-reference	RA: 01 29 36.1357 (22.4005654d)				Proper Motion RA: 56.46899999999994 mas/yr						
			Dec: -21 37 45.61 (-21.62934d)				Proper Motion Dec: 4.134 mas/yr						
			Equinox: J2000				Parallax: 0.013896499999999999"						
							Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[A stars]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	1	2	0.0		-16.7		0.0		10.0		DEFAULT		
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		9					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2		

Proposal 7313 - Observation 2 - Determining the Origin of the Gas in the 49 Ceti Debris Disk

Special Requirements	Aperture PA Range 201.2 to 221.2 Degrees (V3 62.22835083 to 82.22835083) Group Observations 1, 2, Non-interruptible
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